STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

Marmara University **Faculty of Engineering**	**EE2001 and EE201 Make-up Exam**	**19.02.2021** **PAGE 1/4**

PLEASE FIRST READ HERE
General Instructions:

- The exam starts at **11.00** and ends at **12.45**. You have **105 MINUTES**. There won't be any extra time.
- The file upload deadline is **12.45 TODAY** (**19 February 2021**).
- Please devote **15 minutes to file upload** and **ensure that you uploaded everything until 12.45**. You **can not upload or send any files after 12.45**.

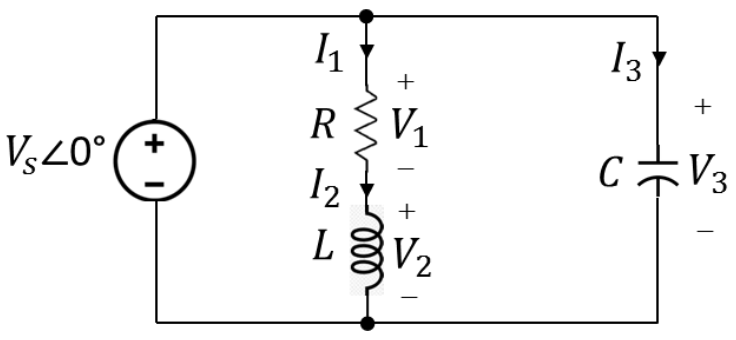
- This exam has 4 questions, each on a different page.
- Please **SOLVE THE QUESTIONS WITH YOUR HAND ON EMPTY PAPERS**, write your **name**, **surname** and **signature** to the top of each paper, take their photos or scan them and upload to UZEM-UES system. Please **write each step of your solution** clearly.
- After each question, a Table including the student specific question parameters is provided. Please find the row with your student number in each table and use the parameters in that row when solving the question.
- If you face any problems or would like to ask any questions about the exam, please send e-mail to: alparslan.zehir@marmara.edu.tr

QUESTION 1/4:

Note: In this question, for $\boldsymbol{V_s}$, $\boldsymbol{R}$, $\boldsymbol{L}$ and $\boldsymbol{C}$, please use the **numerical values and units** provided for you in the same row with your student number in Table 1 in the next page.

The circuit given below has a **sinusoidal** voltage source with $\boldsymbol{V_s \angle 0°}$ V rms voltage and $\boldsymbol{\omega = 4000\ rad/s}$, one resistor with $\boldsymbol{R}$ Ω, an inductor with $\boldsymbol{L}$ mH and a capacitor with $\boldsymbol{C}$ μF. Please answer the following:

(25 pts.) For each passive element in the circuit, find the rms phasor voltages (V) and currents (I) together with the real powers (P) and magnetizing reactive powers (Q). Please **write each step of your calculations in your solution clearly**. You can write your final results in a table similar to the one given below.



An **example table format** to provide your results, **after** presenting your calculations.

	Resistor	Capacitor	Inductor
$V\ (Volts\ rms)$			
$I\ (Ampers\ rms)$			
$P\ (Watts)$			
$Q\ (VAr)$			

Table 1. Question 1 Parameters

Student No	Student Surname	QUESTION 1 Parameters			
		V_s (V)	R (Ω)	L (mH)	C (μF)
100714004	ATLIHAN	4	12	20	6
150116063	OKUMUŞ	8	24	40	12
150712026	TÜRKMEN	12	36	60	18
150712030	ERMAN	16	48	80	24
150712049	SALTABAŞ	20	60	100	30
150713030	AKBAŞ	24	72	120	36
150714020	AYDIN	28	84	140	42
150714021	AKÖZ	32	96	160	48
150714023	BOSTANCIOĞLU	36	108	180	54
150714048	AYDIN	40	120	200	60
150715037	PEKTAŞ	44	132	220	66
150715044	ASLAN	48	144	240	72
150715046	UYSAL	52	156	260	78
150715050	YILDIZ	56	168	280	84
150715052	TANER	60	180	300	90
150715053	SERTLEROĞULLARI	64	192	320	96
150716001	DAĞDAŞ	68	204	340	102
150716009	YÜKCÜ	72	216	360	108
150716016	ERYILMAZ	76	228	380	114
150716024	YILMAZ	80	240	400	120
150716026	KAHRAMAN	84	252	420	126
150716028	KURMUŞ	88	264	440	132
150716030	AKDEMİR	92	276	460	138
150716033	BİLGİÇ	96	288	480	144
150716036	CÜCE	100	300	500	150
150716039	ÇOKLUK	104	312	520	156
150716043	HALIGÜR	108	324	540	162
150716046	KARACA	112	336	560	168
150716057	ULU	116	348	580	174
150716075	KARTAL	120	360	600	180
150716501	KAHYA	124	372	620	186
150716502	ÖZTÜRK	128	384	640	192
150717001	ATAN	132	396	660	198
150717004	DAĞLI	136	408	680	204
150717005	CANSIZER	140	420	700	210
150717008	BUĞUR	144	432	720	216
150717009	EKİCİ	148	444	740	222
150717010	ÖZÇELİK	152	456	760	228
150717014	MOR	156	468	780	234
150717015	AKTAŞ	160	480	800	240
150717019	BİNGÖL	164	492	820	246
150717021	AVCIOĞLU	168	504	840	252
150717022	ÖZCAN	172	516	860	258
150717024	GÖNÜL	176	528	880	264

Student No	Student Surname	QUESTION 1 Parameters			
		V_s (V)	R (Ω)	L (mH)	C (μF)
150717025	CANBAZ	180	540	900	270
150717031	ÖZTÜRK	184	552	920	276
150717032	BALCI	188	564	940	282
150717034	ÖZKUL	192	576	960	288
150717036	VURUCU	196	588	980	294
150717037	YILMAZ	200	600	1000	300
150717038	DUMLU	204	612	1020	306
150717043	TÜRK	208	624	1040	312
150717047	ŞAHİN	212	636	1060	318
150717049	ÖZKAN	216	648	1080	324
150717051	PATOĞLU	220	660	1100	330
150717054	ARSLAN	224	672	1120	336
150717055	TEMİZ	228	684	1140	342
150717058	ÇÖMEZ	232	696	1160	348
150717060	TEKELİ	236	708	1180	354
150717853	KOCABAŞ	240	720	1200	360
150718003	BOYRAZ	244	732	1220	366
150718005	ELBİR	248	744	1240	372
150718006	SEZGİN	252	756	1260	378
150718008	KARAGÖZLÜ	256	768	1280	384
150718010	GÜNDOĞAN	260	780	1300	390
150718011	YURDAKUL	264	792	1320	396
150718014	DİŞİBÜYÜK	268	804	1340	402
150718015	BÜYÜKKAĞNICI	272	816	1360	408
150718016	TÜRK	276	828	1380	414
150718017	ŞAHİN	280	840	1400	420
150718018	DELİASLAN	284	852	1420	426
150718019	ABDİ	20	6	1800	4
150718020	ŞAHİN	24	18	1820	10
150718021	GÜNDÜZ	28	30	1840	16
150718022	BEDİR	32	42	1860	22
150718023	YAVAŞ	36	54	1880	28
150718024	KARASLAN	40	66	1900	34
150718026	MERT	44	78	1920	40
150718028	UZUNER	48	90	1940	46
150718029	AYDOĞAN	52	102	1960	52
150718031	EROĞLU	56	114	1980	58
150718033	TEMİR	60	126	2000	64
150718034	ER	64	138	2020	70
150718035	AKYÜREK	68	150	2040	76
150718036	SARICAN	72	162	2060	82
150718037	ATABAY	76	174	2080	88
150718039	KÜÇÜKKÖŞKER	80	186	2100	94
150718042	AĞCABAY	84	198	2120	100
150718043	TİRYAKİ	88	210	2140	106
150718045	KALKIR	92	222	2160	112

Student No	Student Surname	QUESTION 1 Parameters			
		V_s (V)	R (Ω)	L (mH)	C (μF)
150718046	TUNÇBİLEK	96	234	2180	118
150718047	MANZAK	100	246	2200	124
150718048	KAYA	104	258	2220	130
150718049	GÜÇ	108	270	2240	136
150718050	KARACA	112	282	2260	142
150718051	EROĞLU	116	294	2280	148
150718052	TOK	120	306	2300	154
150718053	BAYRAM	124	318	2320	160
150718055	BEBEK	128	330	2340	166
150718056	CAFEROĞLU	132	342	2360	172
150718058	ÖZTOPRAK	136	354	2380	178
150718059	SÜNE	140	366	2400	184
150718060	DOĞAN	144	378	2420	190
150718061	TERZİ	148	390	2440	196
150718062	ATASEVER	152	402	2460	202
150718501	KİRAZ	156	414	2480	208
150718503	AYDI	160	426	2500	214
150718852	ULU	164	438	2520	220
150719009	NECİP	168	450	2540	226
150719011	DENİZ	172	462	2560	232
150719015	AYDİN	176	474	2580	238
150719016	İBİŞ	180	486	2600	244
150719040	BÜYÜKKAYA	184	498	2620	250
150719054	GÜRBÜZ	188	510	2640	256
150719058	GEDİK	192	522	2660	262
150719518	ÜNLÜ	196	534	2680	268
150719523	UZUN	200	546	2700	274
150719533	AYDIN	204	558	2720	280
150719582	MAROUF	208	570	2740	286
150719583	ALHAJ MASOUD	212	582	2760	292
150719608	OKURER	216	594	2780	298
150719621	SÖKMEN	220	606	2800	304
150719634	ŞAHİN	224	618	2820	310
150719662	LADİKLİOĞLU	228	630	2840	316
150719663	TURAKOĞLU	232	642	2860	322
150719827	FİDAN	236	654	2880	328
150719828	UĞURLUER	240	666	2900	334
150719829	ERDEM	244	678	2920	340
150719841	MAMATİ	248	690	2940	346
150720821	KOÇMARLI	252	702	2960	352
150720822	AYGÜN	256	714	2980	358
150720823	TÜĞMEN	260	726	3000	364
150818005	HACIOĞLU	264	738	3020	370
199520029	VISINI	272	762	3060	382
199819097	ESERÇELİK	276	774	3080	388

STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

QUESTION 2/4:

Note: In this question, for $\boldsymbol{t_1, V_s, R_1, R_2, R_3}$ and $\boldsymbol{L}$, please use the **numerical values and units** provided for you in the same row with your student number in Table 2 in the next page.

The voltage source in the given circuit has DC voltage. The inductor in the given circuit has been connected to terminal d of the three-position switch for a long time. At $t = 0$, the switch is moved to position e, where it remains for $\boldsymbol{t_1}$ **ms**. After that, the switch is moved to position f where it remains indefinitely. Please answer the following:

(15 pts.) 2.a) Using the general solution method, derive the numerical expressions for the voltage across the capacitor when the switch was moved to terminals e and f (one expression for each switch position).

(10 pts.) 2.b) Considering that the initial value of the current in the inductor as $\boldsymbol{I_o}$, when the current in the inductor is $\frac{\boldsymbol{I_o}}{\boldsymbol{2}}$?

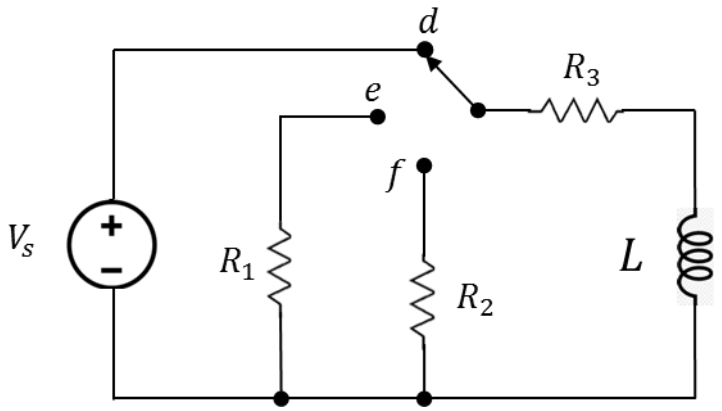


Table 2. Question 2 Parameters

Student No	Student Surname	QUESTION 2 Parameters					
		t_1 (ms)	V_s (V)	R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	L (mH)
100714004	ATLIHAN	15	8	6	4	4	0.4
150116063	OKUMUŞ	30	16	12	8	8	0.8
150712026	TÜRKMEN	45	24	18	12	12	1.2
150712030	ERMAN	60	32	24	16	16	1.6
150712049	SALTABAŞ	75	40	30	20	20	2
150713030	AKBAŞ	90	48	36	24	24	2.4
150714020	AYDIN	105	56	42	28	28	2.8
150714021	AKÖZ	120	64	48	32	32	3.2
150714023	BOSTANCIOĞLU	135	72	54	36	36	3.6
150714048	AYDIN	150	80	60	40	40	4
150715037	PEKTAŞ	165	88	66	44	44	4.4
150715044	ASLAN	180	96	72	48	48	4.8
150715046	UYSAL	195	104	78	52	52	5.2
150715050	YILDIZ	210	112	84	56	56	5.6
150715052	TANER	225	120	90	60	60	6
150715053	SERTLEROĞULLARI	240	128	96	64	64	6.4
150716001	DAĞDAŞ	255	136	102	68	68	6.8
150716009	YÜKCÜ	270	144	108	72	72	7.2
150716016	ERYILMAZ	285	152	114	76	76	7.6
150716024	YILMAZ	300	160	120	80	80	8
150716026	KAHRAMAN	315	168	126	84	84	8.4
150716028	KURMUŞ	330	176	132	88	88	8.8
150716030	AKDEMİR	345	184	138	92	92	9.2
150716033	BİLGİÇ	360	192	144	96	96	9.6
150716036	CÜCE	375	200	150	100	100	10
150716039	ÇOKLUK	390	208	156	104	104	10.4
150716043	HALIGÜR	405	216	162	108	108	10.8
150716046	KARACA	420	224	168	112	112	11.2
150716057	ULU	435	232	174	116	116	11.6
150716075	KARTAL	450	240	180	120	120	12
150716501	KAHYA	465	248	186	124	124	12.4
150716502	ÖZTÜRK	480	256	192	128	128	12.8
150717001	ATAN	495	264	198	132	132	13.2
150717004	DAĞLI	510	272	204	136	136	13.6
150717005	CANSIZER	525	280	210	140	140	14
150717008	BUĞUR	540	288	216	144	144	14.4
150717009	EKİCİ	555	296	222	148	148	14.8
150717010	ÖZÇELİK	570	304	228	152	152	15.2
150717014	MOR	585	312	234	156	156	15.6
150717015	AKTAŞ	600	320	240	160	160	16
150717019	BİNGÖL	615	328	246	164	164	16.4
150717021	AVCIOĞLU	630	336	252	168	168	16.8
150717022	ÖZCAN	645	344	258	172	172	17.2
150717024	GÖNÜL	660	352	264	176	176	17.6

Student No	Student Surname	QUESTION 2 Parameters					
		t_1 (ms)	V_s (V)	R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	C (μF)
150717025	CANBAZ	675	360	270	180	180	18
150717031	ÖZTÜRK	690	368	276	184	184	18.4
150717032	BALCI	705	376	282	188	188	18.8
150717034	ÖZKUL	720	384	288	192	192	19.2
150717036	VURUCU	735	392	294	196	196	19.6
150717037	YILMAZ	750	400	300	200	200	20
150717038	DUMLU	765	408	306	204	204	20.4
150717043	TÜRK	780	416	312	208	208	20.8
150717047	ŞAHİN	795	424	318	212	212	21.2
150717049	ÖZKAN	810	432	324	216	216	21.6
150717051	PATOĞLU	825	440	330	220	220	22
150717054	ARSLAN	840	448	336	224	224	22.4
150717055	TEMİZ	855	456	342	228	228	22.8
150717058	ÇÖMEZ	870	464	348	232	232	23.2
150717060	TEKELİ	885	472	354	236	236	23.6
150717853	KOCABAŞ	900	480	360	240	240	24
150718003	BOYRAZ	915	488	366	244	244	24.4
150718005	ELBİR	930	496	372	248	248	24.8
150718006	SEZGİN	945	504	378	252	252	25.2
150718008	KARAGÖZLÜ	960	512	384	256	256	25.6
150718010	GÜNDOĞAN	975	520	390	260	260	26
150718011	YURDAKUL	990	528	396	264	264	26.4
150718014	DİŞİBÜYÜK	1005	536	402	268	268	26.8
150718015	BÜYÜKKAĞNICI	1020	544	408	272	272	27.2
150718016	TÜRK	1035	552	414	276	276	27.6
150718017	ŞAHİN	1050	560	420	280	280	28
150718018	DELİASLAN	1065	568	426	284	284	28.4
150718019	ABDİ	5	50	20	10	6	0.4
150718020	ŞAHİN	20	58	26	14	10	0.8
150718021	GÜNDÜZ	35	66	32	18	14	1.2
150718022	BEDİR	50	74	38	22	18	1.6
150718023	YAVAŞ	65	82	44	26	22	2
150718024	KARASLAN	80	90	50	30	26	2.4
150718026	MERT	95	98	56	34	30	2.8
150718028	UZUNER	110	106	62	38	34	3.2
150718029	AYDOĞAN	125	114	68	42	38	3.6
150718031	EROĞLU	140	122	74	46	42	4
150718033	TEMİR	155	130	80	50	46	4.4
150718034	ER	170	138	86	54	50	4.8
150718035	AKYÜREK	185	146	92	58	54	5.2
150718036	SARICAN	200	154	98	62	58	5.6
150718037	ATABAY	215	162	104	66	62	6
150718039	KÜÇÜKKÖŞKER	230	170	110	70	66	6.4
150718042	AĞCABAY	245	178	116	74	70	6.8
150718043	TİRYAKİ	260	186	122	78	74	7.2
150718045	KALKIR	275	194	128	82	78	7.6

Student No	Student Surname	QUESTION 2 Parameters					
		t_1 (ms)	V_s (V)	R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	C (μF)
150718046	TUNÇBİLEK	290	202	134	86	82	8
150718047	MANZAK	305	210	140	90	86	8.4
150718048	KAYA	320	218	146	94	90	8.8
150718049	GÜÇ	335	226	152	98	94	9.2
150718050	KARACA	350	234	158	102	98	9.6
150718051	EROĞLU	365	242	164	106	102	10
150718052	TOK	380	250	170	110	106	10.4
150718053	BAYRAM	395	258	176	114	110	10.8
150718055	BEBEK	410	266	182	118	114	11.2
150718056	CAFEROĞLU	425	274	188	122	118	11.6
150718058	ÖZTOPRAK	440	282	194	126	122	12
150718059	SÜNE	455	290	200	130	126	12.4
150718060	DOĞAN	470	298	206	134	130	12.8
150718061	TERZİ	485	306	212	138	134	13.2
150718062	ATASEVER	500	314	218	142	138	13.6
150718501	KİRAZ	515	322	224	146	142	14
150718503	AYDI	530	330	230	150	146	14.4
150718852	ULU	545	338	236	154	150	14.8
150719009	NECİP	560	346	242	158	154	15.2
150719011	DENİZ	575	354	248	162	158	15.6
150719015	AYDİN	590	362	254	166	162	16
150719016	İBİŞ	605	370	260	170	166	16.4
150719040	BÜYÜKKAYA	620	378	266	174	170	16.8
150719054	GÜRBÜZ	635	386	272	178	174	17.2
150719058	GEDİK	650	394	278	182	178	17.6
150719518	ÜNLÜ	665	402	284	186	182	18
150719523	UZUN	680	410	290	190	186	18.4
150719533	AYDIN	695	418	296	194	190	18.8
150719582	MAROUF	710	426	302	198	194	19.2
150719583	ALHAJ MASOUD	725	434	308	202	198	19.6
150719608	OKURER	740	442	314	206	202	20
150719621	SÖKMEN	755	450	320	210	206	20.4
150719634	ŞAHİN	770	458	326	214	210	20.8
150719662	LADİKLİOĞLU	785	466	332	218	214	21.2
150719663	TURAKOĞLU	800	474	338	222	218	21.6
150719827	FİDAN	815	482	344	226	222	22
150719828	UĞURLUER	830	490	350	230	226	22.4
150719829	ERDEM	845	498	356	234	230	22.8
150719841	MAMATİ	860	506	362	238	234	23.2
150720821	KOÇMARLI	875	514	368	242	238	23.6
150720822	AYGÜN	890	522	374	246	242	24
150720823	TÜĞMEN	905	530	380	250	246	24.4
150818005	HACIOĞLU	920	538	386	254	250	24.8
199520029	VISINI	950	554	398	262	258	25.6
199819097	ESERÇELİK	965	562	404	266	262	26

STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

QUESTION 3/4:

Note: In this question, for $\boldsymbol{R_1, jX_1, V_L,}$ and $\boldsymbol{S_L,}$ please use the **numerical values and units** provided for you in the same row with your student number in Table 3 in the next page.

Load 1 absorbs a complex power of $\boldsymbol{S_L}$ **VA** at a **leading** power factor of 0.6.
Load 2 absorbs a complex power of $\boldsymbol{S_L}$ **VA** at a **lagging** power factor of 0.4.

(15 pts.) 3.a) Find $\boldsymbol{V_S}$ (rms) and $\boldsymbol{\theta}$ for the voltage source.

(10 pts.) 3.b) Construct the phasor diagram showing all the voltage and current phasors.

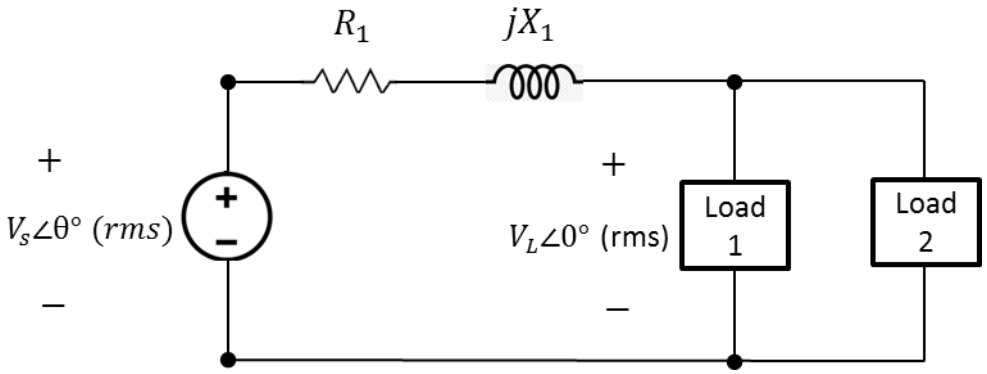


Table 3. Question 3 Parameters

Student No	Student Surname	QUESTION 3 Parameters			
		R_1 (Ω)	jX_1 (Ω)	V_L (V)	S_L (VA)
100714004	ATLIHAN	4	6	8	2
150116063	OKUMUŞ	8	12	16	4
150712026	TÜRKMEN	12	18	24	6
150712030	ERMAN	16	24	32	8
150712049	SALTABAŞ	20	30	40	10
150713030	AKBAŞ	24	36	48	12
150714020	AYDIN	28	42	56	14
150714021	AKÖZ	32	48	64	16
150714023	BOSTANCIOĞLU	36	54	72	18
150714048	AYDIN	40	60	80	20
150715037	PEKTAŞ	44	66	88	22
150715044	ASLAN	48	72	96	24
150715046	UYSAL	52	78	104	26
150715050	YILDIZ	56	84	112	28
150715052	TANER	60	90	120	30
150715053	SERTLEROĞULLARI	64	96	128	32
150716001	DAĞDAŞ	68	102	136	34
150716009	YÜKCÜ	72	108	144	36
150716016	ERYILMAZ	76	114	152	38
150716024	YILMAZ	80	120	160	40
150716026	KAHRAMAN	84	126	168	42
150716028	KURMUŞ	88	132	176	44
150716030	AKDEMİR	92	138	184	46
150716033	BİLGİÇ	96	144	192	48
150716036	CÜCE	100	150	200	50
150716039	ÇOKLUK	104	156	208	52
150716043	HALIGÜR	108	162	216	54
150716046	KARACA	112	168	224	56
150716057	ULU	116	174	232	58
150716075	KARTAL	120	180	240	60
150716501	KAHYA	124	186	248	62
150716502	ÖZTÜRK	128	192	256	64
150717001	ATAN	132	198	264	66
150717004	DAĞLI	136	204	272	68
150717005	CANSIZER	140	210	280	70
150717008	BUĞUR	144	216	288	72
150717009	EKİCİ	148	222	296	74
150717010	ÖZÇELİK	152	228	304	76
150717014	MOR	156	234	312	78
150717015	AKTAŞ	160	240	320	80
150717019	BİNGÖL	164	246	328	82
150717021	AVCIOĞLU	168	252	336	84
150717022	ÖZCAN	172	258	344	86
150717024	GÖNÜL	176	264	352	88

Student No	Student Surname	QUESTION 3 Parameters			
		R_1 (Ω)	jX_1 (Ω)	V_L (V)	S_L (VA)
150717025	CANBAZ	180	270	360	90
150717031	ÖZTÜRK	184	276	368	92
150717032	BALCI	188	282	376	94
150717034	ÖZKUL	192	288	384	96
150717036	VURUCU	196	294	392	98
150717037	YILMAZ	200	300	400	100
150717038	DUMLU	204	306	408	102
150717043	TÜRK	208	312	416	104
150717047	ŞAHİN	212	318	424	106
150717049	ÖZKAN	216	324	432	108
150717051	PATOĞLU	220	330	440	110
150717054	ARSLAN	224	336	448	112
150717055	TEMİZ	228	342	456	114
150717058	ÇÖMEZ	232	348	464	116
150717060	TEKELİ	236	354	472	118
150717853	KOCABAŞ	240	360	480	120
150718003	BOYRAZ	244	366	488	122
150718005	ELBİR	248	372	496	124
150718006	SEZGİN	252	378	504	126
150718008	KARAGÖZLÜ	256	384	512	128
150718010	GÜNDOĞAN	260	390	520	130
150718011	YURDAKUL	264	396	528	132
150718014	DİŞİBÜYÜK	268	402	536	134
150718015	BÜYÜKKAĞNICI	272	408	544	136
150718016	TÜRK	276	414	552	138
150718017	ŞAHİN	280	420	560	140
150718018	DELİASLAN	284	426	568	142
150718019	ABDİ	6	14	10	4
150718020	ŞAHİN	10	20	18	6
150718021	GÜNDÜZ	14	26	26	8
150718022	BEDİR	18	32	34	10
150718023	YAVAŞ	22	38	42	12
150718024	KARASLAN	26	44	50	14
150718026	MERT	30	50	58	16
150718028	UZUNER	34	56	66	18
150718029	AYDOĞAN	38	62	74	20
150718031	EROĞLU	42	68	82	22
150718033	TEMİR	46	74	90	24
150718034	ER	50	80	98	26
150718035	AKYÜREK	54	86	106	28
150718036	SARICAN	58	92	114	30
150718037	ATABAY	62	98	122	32
150718039	KÜÇÜKKÖŞKER	66	104	130	34
150718042	AĞCABAY	70	110	138	36
150718043	TİRYAKİ	74	116	146	38

Student No	Student Surname	QUESTION 3 Parameters			
		R_1 (Ω)	jX_1 (Ω)	V_L (V)	S_L (VA)
150718045	KALKIR	78	122	154	40
150718046	TUNÇBİLEK	82	128	162	42
150718047	MANZAK	86	134	170	44
150718048	KAYA	90	140	178	46
150718049	GÜÇ	94	146	186	48
150718050	KARACA	98	152	194	50
150718051	EROĞLU	102	158	202	52
150718052	TOK	106	164	210	54
150718053	BAYRAM	110	170	218	56
150718055	BEBEK	114	176	226	58
150718056	CAFEROĞLU	118	182	234	60
150718058	ÖZTOPRAK	122	188	242	62
150718059	SÜNE	126	194	250	64
150718060	DOĞAN	130	200	258	66
150718061	TERZİ	134	206	266	68
150718062	ATASEVER	138	212	274	70
150718501	KİRAZ	142	218	282	72
150718503	AYDI	146	224	290	74
150718852	ULU	150	230	298	76
150719009	NECİP	154	236	306	78
150719011	DENİZ	158	242	314	80
150719015	AYDİN	162	248	322	82
150719016	İBİŞ	166	254	330	84
150719040	BÜYÜKKAYA	170	260	338	86
150719054	GÜRBÜZ	174	266	346	88
150719058	GEDİK	178	272	354	90
150719518	ÜNLÜ	182	278	362	92
150719523	UZUN	186	284	370	94
150719533	AYDIN	190	290	378	96
150719582	MAROUF	194	296	386	98
150719583	ALHAJ MASOUD	198	302	394	100
150719608	OKURER	202	308	402	102
150719621	SÖKMEN	206	314	410	104
150719634	ŞAHİN	210	320	418	106
150719662	LADİKLİOĞLU	214	326	426	108
150719663	TURAKOĞLU	218	332	434	110
150719827	FİDAN	222	338	442	112
150719828	UĞURLUER	226	344	450	114
150719829	ERDEM	230	350	458	116
150719841	MAMATİ	234	356	466	118
150720821	KOÇMARLI	238	362	474	120
150720822	AYGÜN	242	368	482	122
150720823	TÜĞMEN	246	374	490	124
150818005	HACIOĞLU	250	380	498	126
199520029	VISINI	258	392	514	130
199819097	ESERÇELİK	262	398	522	132

STUDENT NAME AND SURNAME :_____

STUDENT SIGNATURE :_____

QUESTION 4/4:

Note: In this question, for v_a, v_b, R_1, R_2, R_3, R_4 and V_{CC}, please use the **numerical values and units** provided for you in the same row with your student number in Table 4 in the next page.

The op-amp in the given circuit is ideal. The input voltages and the supply voltages are all DC voltages. Please answer the following:

(15 pts.) 4.a) Calcuate v_o.

(10 pts.) 4.b) If v_b=**2.4 V,** specify the range of v_a such that the amplifier does not saturate.

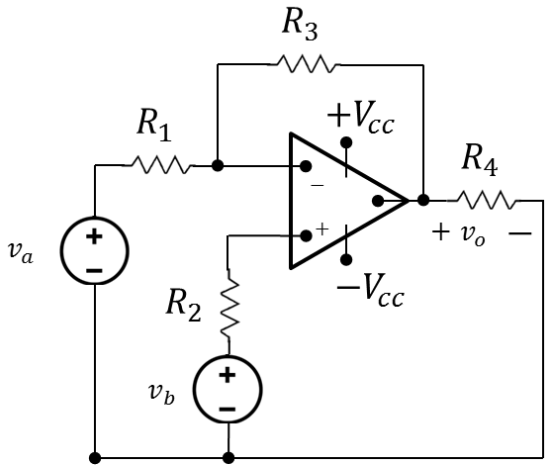


Table 4. Question 4 Parameters

Student No	Student Surname	QUESTION 4 Parameters						
		v_a (V)	v_b (V)	R_1 (Ω)	R_2 (Ω)	R_3 (Ω)	R_4 (Ω)	V_{cc} (V)
100714004	ATLIHAN	6	2	10	20	40	30	3
150116063	OKUMUŞ	12	4	20	40	80	60	6
150712026	TÜRKMEN	18	6	30	60	120	90	9
150712030	ERMAN	24	8	40	80	160	120	12
150712049	SALTABAŞ	30	10	50	100	200	150	15
150713030	AKBAŞ	36	12	60	120	240	180	18
150714020	AYDIN	42	14	70	140	280	210	21
150714021	AKÖZ	48	16	80	160	320	240	24
150714023	BOSTANCIOĞLU	54	18	90	180	360	270	27
150714048	AYDIN	60	20	100	200	400	300	30
150715037	PEKTAŞ	66	22	110	220	440	330	33
150715044	ASLAN	72	24	120	240	480	360	36
150715046	UYSAL	78	26	130	260	520	390	39
150715050	YILDIZ	84	28	140	280	560	420	42
150715052	TANER	90	30	150	300	600	450	45
150715053	SERTLEROĞULLARI	96	32	160	320	640	480	48
150716001	DAĞDAŞ	102	34	170	340	680	510	51
150716009	YÜKCÜ	108	36	180	360	720	540	54
150716016	ERYILMAZ	114	38	190	380	760	570	57
150716024	YILMAZ	120	40	200	400	800	600	60
150716026	KAHRAMAN	126	42	210	420	840	630	63
150716028	KURMUŞ	132	44	220	440	880	660	66
150716030	AKDEMİR	138	46	230	460	920	690	69
150716033	BİLGİÇ	144	48	240	480	960	720	72
150716036	CÜCE	150	50	250	500	1000	750	75
150716039	ÇOKLUK	156	52	260	520	1040	780	78
150716043	HALIGÜR	162	54	270	540	1080	810	81
150716046	KARACA	168	56	280	560	1120	840	84
150716057	ULU	174	58	290	580	1160	870	87
150716075	KARTAL	180	60	300	600	1200	900	90
150716501	KAHYA	186	62	310	620	1240	930	93
150716502	ÖZTÜRK	192	64	320	640	1280	960	96
150717001	ATAN	198	66	330	660	1320	990	99
150717004	DAĞLI	204	68	340	680	1360	1020	102
150717005	CANSIZER	210	70	350	700	1400	1050	105
150717008	BUĞUR	216	72	360	720	1440	1080	108
150717009	EKİCİ	222	74	370	740	1480	1110	111
150717010	ÖZÇELİK	228	76	380	760	1520	1140	114
150717014	MOR	234	78	390	780	1560	1170	117
150717015	AKTAŞ	240	80	400	800	1600	1200	120
150717019	BİNGÖL	246	82	410	820	1640	1230	123
150717021	AVCIOĞLU	252	84	420	840	1680	1260	126
150717022	ÖZCAN	258	86	430	860	1720	1290	129
150717024	GÖNÜL	264	88	440	880	1760	1320	132

Student No	Student Surname	QUESTION 4 Parameters						
		v_a (V)	v_b (V)	R_1 (Ω)	R_2 (Ω)	R_3 (Ω)	R_4 (Ω)	V_{cc} (V)
150717025	CANBAZ	270	90	450	900	1800	1350	135
150717031	ÖZTÜRK	276	92	460	920	1840	1380	138
150717032	BALCI	282	94	470	940	1880	1410	141
150717034	ÖZKUL	288	96	480	960	1920	1440	144
150717036	VURUCU	294	98	490	980	1960	1470	147
150717037	YILMAZ	300	100	500	1000	2000	1500	150
150717038	DUMLU	306	102	510	1020	2040	1530	153
150717043	TÜRK	312	104	520	1040	2080	1560	156
150717047	ŞAHİN	318	106	530	1060	2120	1590	159
150717049	ÖZKAN	324	108	540	1080	2160	1620	162
150717051	PATOĞLU	330	110	550	1100	2200	1650	165
150717054	ARSLAN	336	112	560	1120	2240	1680	168
150717055	TEMİZ	342	114	570	1140	2280	1710	171
150717058	ÇÖMEZ	348	116	580	1160	2320	1740	174
150717060	TEKELİ	354	118	590	1180	2360	1770	177
150717853	KOCABAŞ	360	120	600	1200	2400	1800	180
150718003	BOYRAZ	366	122	610	1220	2440	1830	183
150718005	ELBİR	372	124	620	1240	2480	1860	186
150718006	SEZGİN	378	126	630	1260	2520	1890	189
150718008	KARAGÖZLÜ	384	128	640	1280	2560	1920	192
150718010	GÜNDOĞAN	390	130	650	1300	2600	1950	195
150718011	YURDAKUL	396	132	660	1320	2640	1980	198
150718014	DİŞİBÜYÜK	402	134	670	1340	2680	2010	201
150718015	BÜYÜKKAĞNICI	408	136	680	1360	2720	2040	204
150718016	TÜRK	414	138	690	1380	2760	2070	207
150718017	ŞAHİN	420	140	700	1400	2800	2100	210
150718018	DELİASLAN	426	142	710	1420	2840	2130	213
150718019	ABDİ	4	0	20	40	100	50	15
150718020	ŞAHİN	10	2	30	60	140	80	18
150718021	GÜNDÜZ	16	4	40	80	180	110	21
150718022	BEDİR	22	6	50	100	220	140	24
150718023	YAVAŞ	28	8	60	120	260	170	27
150718024	KARASLAN	34	10	70	140	300	200	30
150718026	MERT	40	12	80	160	340	230	33
150718028	UZUNER	46	14	90	180	380	260	36
150718029	AYDOĞAN	52	16	100	200	420	290	39
150718031	EROĞLU	58	18	110	220	460	320	42
150718033	TEMİR	64	20	120	240	500	350	45
150718034	ER	70	22	130	260	540	380	48
150718035	AKYÜREK	76	24	140	280	580	410	51
150718036	SARICAN	82	26	150	300	620	440	54
150718037	ATABAY	88	28	160	320	660	470	57
150718039	KÜÇÜKKÖŞKER	94	30	170	340	700	500	60
150718042	AĞCABAY	100	32	180	360	740	530	63
150718043	TİRYAKİ	106	34	190	380	780	560	66

Student No	Student Surname	QUESTION 4 Parameters						
		v_a (V)	v_b (V)	R_1 (Ω)	R_2 (Ω)	R_3 (Ω)	R_4 (Ω)	V_{cc} (V)
150718045	KALKIR	112	36	200	400	820	590	69
150718046	TUNÇBİLEK	118	38	210	420	860	620	72
150718047	MANZAK	124	40	220	440	900	650	75
150718048	KAYA	130	42	230	460	940	680	78
150718049	GÜÇ	136	44	240	480	980	710	81
150718050	KARACA	142	46	250	500	1020	740	84
150718051	EROĞLU	148	48	260	520	1060	770	87
150718052	TOK	154	50	270	540	1100	800	90
150718053	BAYRAM	160	52	280	560	1140	830	93
150718055	BEBEK	166	54	290	580	1180	860	96
150718056	CAFEROĞLU	172	56	300	600	1220	890	99
150718058	ÖZTOPRAK	178	58	310	620	1260	920	102
150718059	SÜNE	184	60	320	640	1300	950	105
150718060	DOĞAN	190	62	330	660	1340	980	108
150718061	TERZİ	196	64	340	680	1380	1010	111
150718062	ATASEVER	202	66	350	700	1420	1040	114
150718501	KİRAZ	208	68	360	720	1460	1070	117
150718503	AYDI	214	70	370	740	1500	1100	120
150718852	ULU	220	72	380	760	1540	1130	123
150719009	NECİP	226	74	390	780	1580	1160	126
150719011	DENİZ	232	76	400	800	1620	1190	129
150719015	AYDİN	238	78	410	820	1660	1220	132
150719016	İBİŞ	244	80	420	840	1700	1250	135
150719040	BÜYÜKKAYA	250	82	430	860	1740	1280	138
150719054	GÜRBÜZ	256	84	440	880	1780	1310	141
150719058	GEDİK	262	86	450	900	1820	1340	144
150719518	ÜNLÜ	268	88	460	920	1860	1370	147
150719523	UZUN	274	90	470	940	1900	1400	150
150719533	AYDIN	280	92	480	960	1940	1430	153
150719582	MAROUF	286	94	490	980	1980	1460	156
150719583	ALHAJ MASOUD	292	96	500	1000	2020	1490	159
150719608	OKURER	298	98	510	1020	2060	1520	162
150719621	SÖKMEN	304	100	520	1040	2100	1550	165
150719634	ŞAHİN	310	102	530	1060	2140	1580	168
150719662	LADİKLİOĞLU	316	104	540	1080	2180	1610	171
150719663	TURAKOĞLU	322	106	550	1100	2220	1640	174
150719827	FİDAN	328	108	560	1120	2260	1670	177
150719828	UĞURLUER	334	110	570	1140	2300	1700	180
150719829	ERDEM	340	112	580	1160	2340	1730	183
150719841	MAMATİ	346	114	590	1180	2380	1760	186
150720821	KOÇMARLI	352	116	600	1200	2420	1790	189
150720822	AYGÜN	358	118	610	1220	2460	1820	192
150720823	TÜĞMEN	364	120	620	1240	2500	1850	195
150818005	HACIOĞLU	370	122	630	1260	2540	1880	198
199520029	VISINI	382	126	650	1300	2620	1940	204
199819097	ESERÇELİK	388	128	660	1320	2660	1970	207